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Brac University

Semester: Summer 2024

Course Code: CSE460

VLSI Design

Section: 01-08

Set
01

Assessment: *Midterm Exam*

Duration: 1 hour 30 minutes

Date: 13 July, 2024

Full Marks: 45

Instructions: You MUST answer **Question 1**. You are also required to answer any two of **Questions 2–4**.

■ Question 1 of 4 [CO2] [15 marks]

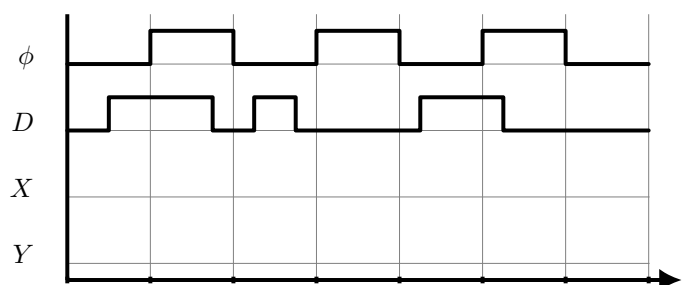
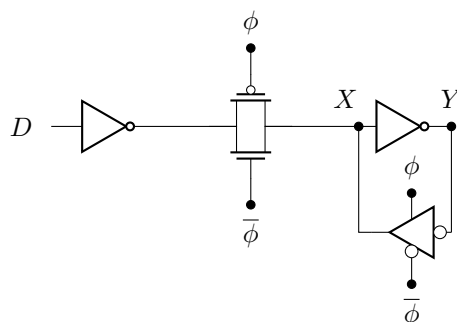
Michael wants to design a Finite State Machine (FSM) that has one input w and one output z . He wants to get $z = 1$ if the previous three values of w were 010; otherwise $z = 0$.

- [5 marks] **Identify** the type of FSM for the above input-output relationship and **design** the state diagram for the machine.
- [5 marks] **Design** the binary-encoded state assigned table of the FSM.
- [5 marks] **Determine** the logic expressions of the next state and output variable z from the state assigned table.

■ Question 2 of 4 [CO1] [15 marks]

There are four sensors, A, B, C, and D that control the lights in a room. Each sensor transitions to a **HIGH** state when it detects an ambient light level exceeding a predefined threshold. The room's lights will turn **ON** when the majority of sensors are in the **LOW** state. If the number of sensors in the **HIGH** and **LOW** states is equal, the room's lighting condition remains **indeterminate** (i.e., a don't care condition). Answer questions (a-c) below based on this description.

- [2 marks] **Write** down the truth table for the system, describing all possible states of the sensors (A, B, C, D) and their corresponding output states.
- [5 marks] **Draw** the Karnaugh map (K-map) for the system based on the truth table in part (a). **Derive** the logical expression that governs the system from the K-map.
- [5 marks] **Draw** the CMOS logic circuit for the logical expression derived in part (b). **Specify** the number of MOSFETs required to implement your design.
- [3 marks] **Complete** the timing diagram shown below on the right, in the question paper, based on the circuit diagram presented below on the left. Ensure that your completed timing diagram accurately reflects the logic behavior of the circuit, taking into account the states of ϕ and D to determine the corresponding outputs X and Y .



■ Question 3 of 4 [CO1] [15 marks]

The parameters for the MOSFETs in the circuits depicted in **Figures 3a** and **3b** are detailed in the **adjacent table**. Additionally, both the pMOS and nMOS devices in these circuits have the same oxide layer thickness, permittivity, and aspect ratio. These are specified as $t_{ox} = 10 \text{ \AA}$, $\epsilon_{ox} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm}$ and $W/L = 50$, respectively.

Parameters for Fig. 3a

nMOS	$\mu_n = 700 \text{ V/cm}^2$	$V_{tn} = 1 \text{ V}$
pMOS	$\mu_p = 350 \text{ V/cm}^2$	$V_{tp} = -1 \text{ V}$

Parameters for Fig. 3b

Both MOSFETs	$\mu_n = 2\mu_p$	$V_{tn} = V_{tp} = 0 \text{ V}$
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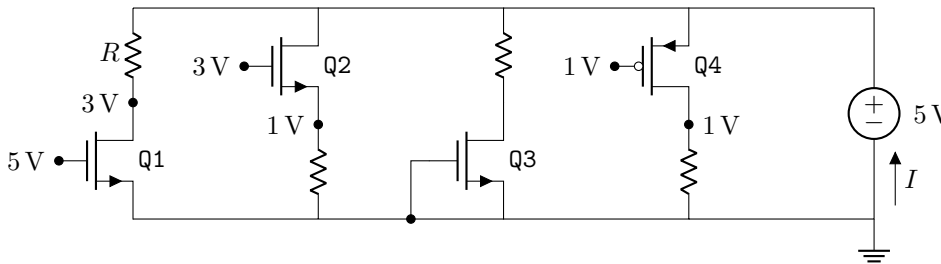


Figure 3a

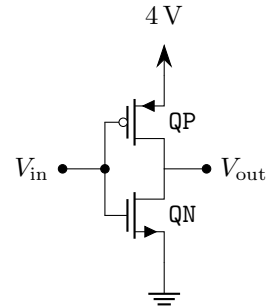


Figure 3b

Answer questions (a-d) based on **Fig. 3a**.

- [2 marks] Find β_n and β_p of the MOSFETs.
- [3 marks] Determine the operating regions of each of the transistors in **Fig. 3a**.
- [3 marks] Find the current I through the 5V voltage source.
- [2 marks] Find the value of R .

Answer the next question based on **Fig. 3b**.

- [5 marks] Find the switchover/inversion voltage V_{inv} of the inverter circuit in **Fig. 3b**.

■ Question 4 of 4 [CO1] [15 marks]

The MOSFET in the circuit of Fig. 2 has the following parameters:

$\mu_p = 225 \text{ V/cm}^2$, $V_t = -0.2 \text{ V}$, $t_{ox} = 10 \text{ \AA}$, $\epsilon_{ox} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm}$ and $W/L = 40$.

- [2 marks] Find β and gate capacitance per unit area of the MOSFET.
- [3 marks] Find the range of V_g for which the MOSFET will be in saturation mode with $I_d = 1 \text{ mA}$.
- [3 marks] What should be an appropriate range of the values for R_1 for the MOSFET to be in saturation mode?
- [2 marks] Find out the operating mode of the MOSFET with proper verification if $R_1 = 3.8 \text{ k}\Omega$, for the lowest V_g value from part (b).

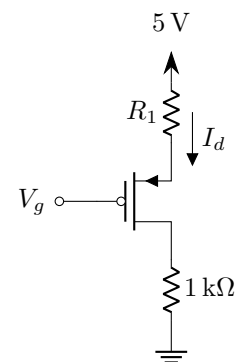


Figure 2

- [5 marks] In the **adjacent figure**, the supply voltage (V_{DD}) is 4V. The threshold voltages for the pMOS and nMOS transistors are both set at $V_{tn} = |V_{tp}| = 0.1 \text{ V}$. Determine the voltages V_1 , V_2 , and V_4 as well as the output voltages V_X and V_Y .

